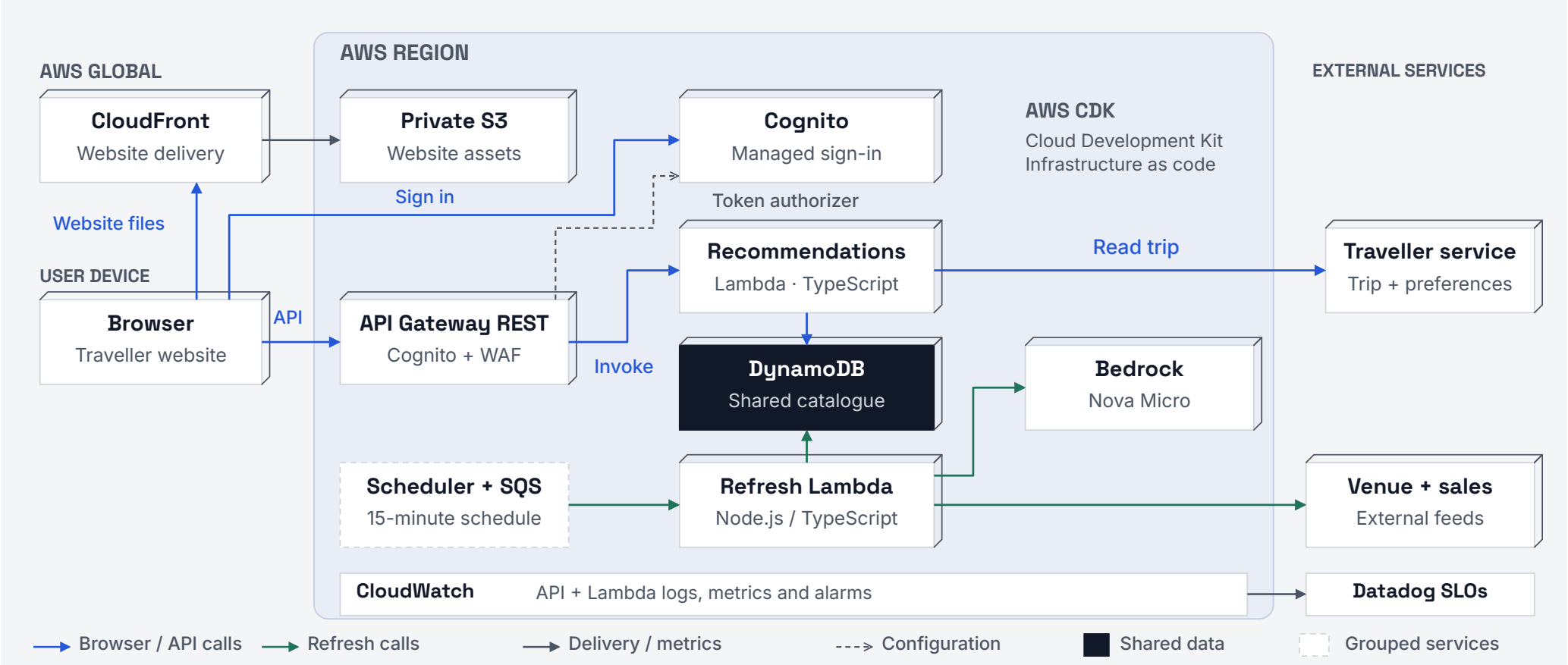
Airport Trip Planner: solution design

The planner recommends airport restaurants and shops using preferences, flight time and sales. A pre-built AI model interprets shop descriptions in the background. Prepared venue information could then be reused for fast recommendations, even when AI is unavailable. But incorrect venue information and estimated visit times can affect the results.

1. AWS infrastructure



Technology choices and trade-offs

Private S3 + CloudFront
Static hosting without servers. **A regional outage interrupts recommendations.**

API Gateway REST, Cognito + WAF
Handles sign-in and protects API access. **Assumes the traveller service accepts Cognito access tokens.**

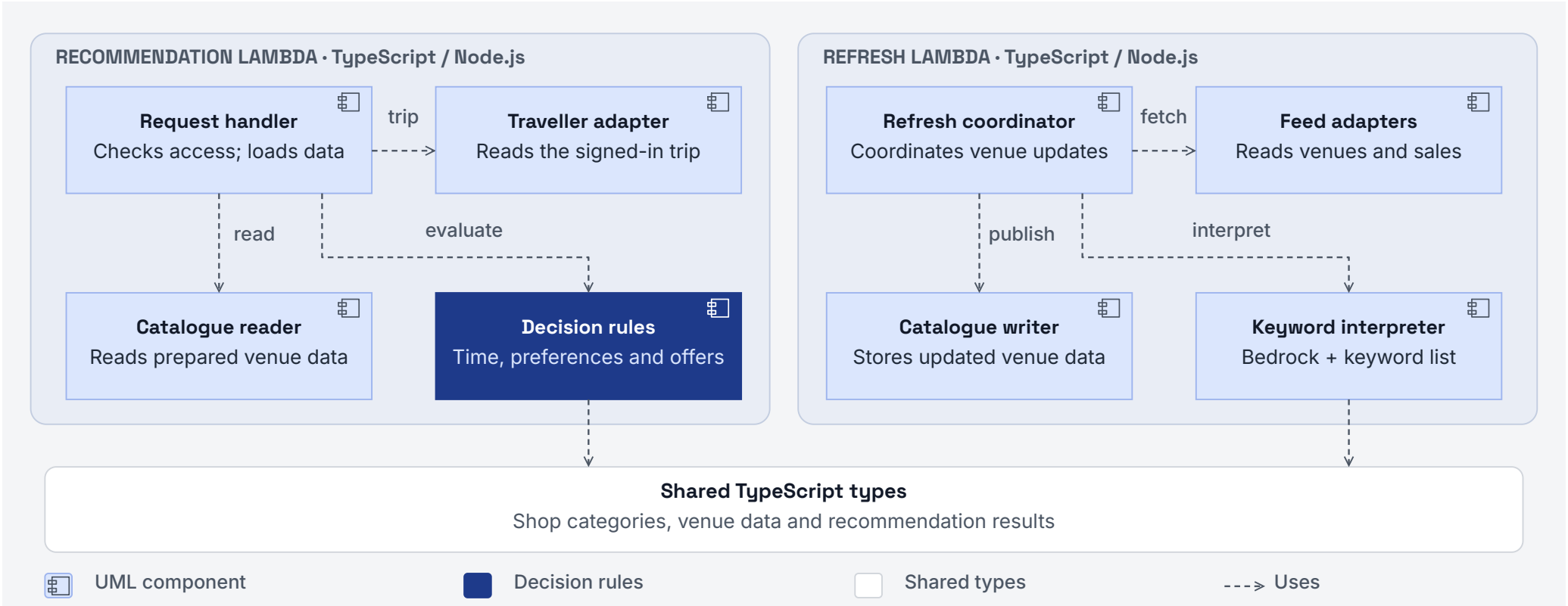
Lambda + DynamoDB
Functions scale separately. Each reads a complete version of the venue data. **Starting a new function instance adds delay.**

EventBridge Scheduler + SQS
Schedules venue updates. **The queue buffers work when updates take longer.**

Bedrock (Nova Micro)
A pre-built AI model interprets the different words brands use to describe their shops. **This avoids training a model, but adds running costs.**

CloudWatch + Datadog
Monitor availability, response time and how current the venue information is.

2. Backend components



Recommendation rules

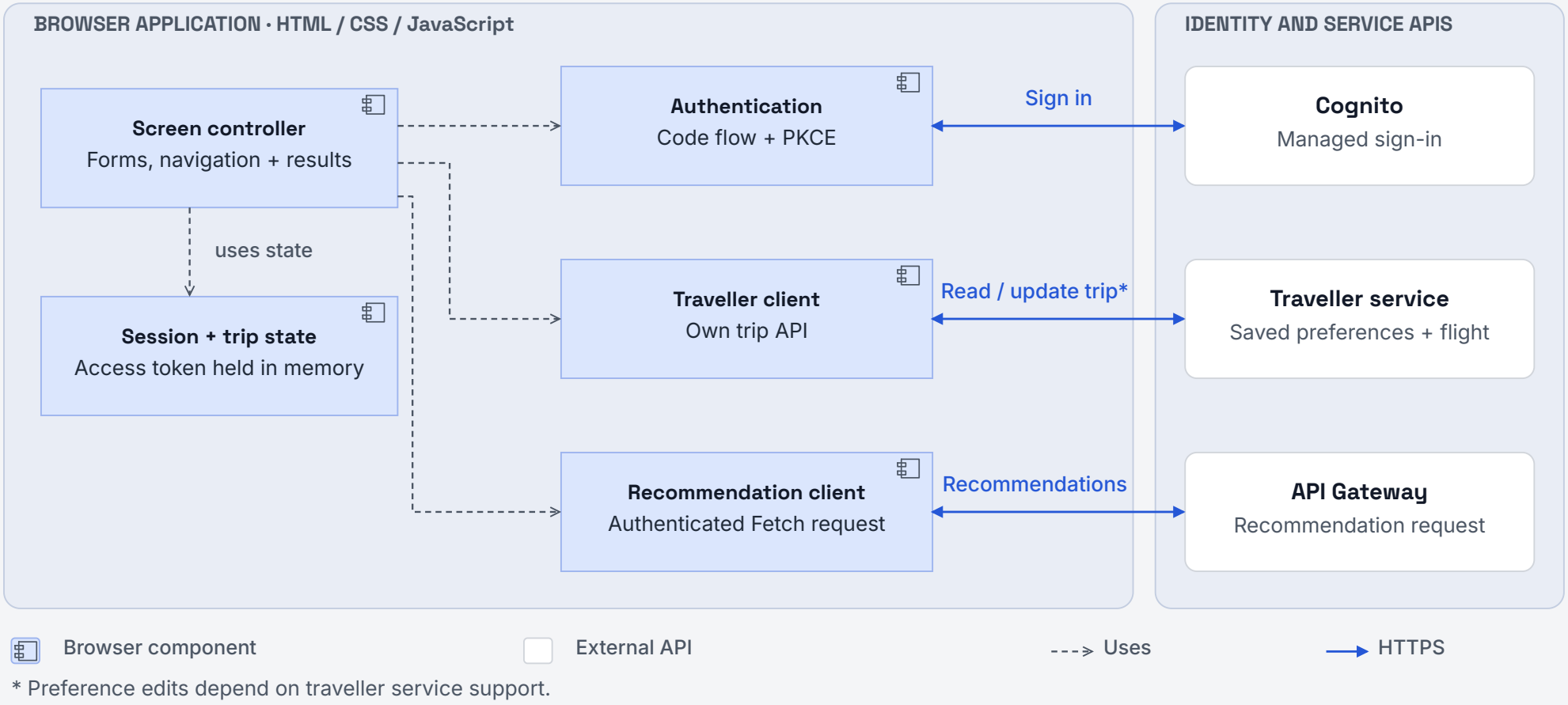
Input	Effect on recommendations
Flight time	Recommend places the traveller has time to visit before boarding.
Preferences	Show matching places first. Dietary requirements use information supplied by the venue.
Sales	Promote valid offers among suitable places. Without preferences, places with offers appear first.
Brand descriptions	Recognise that different words can mean the same thing. For example, “cheap” and “budget” both describe an affordable option.

Technology and quality objectives

TypeScript / Node.js provides shared data types for both functions.

Quality goal	Measure and 30-day design target
Availability	At least 99.9% of valid, authorised requests return recommendations, gate advice or an explanation that no suitable venue is available. Failures in the planner or its dependencies count as unavailable.
Response time	At least 95% of the same requests return a useful result within two seconds, measured at API Gateway. Delays in the traveller service affect this target.
Current venue information	At least 99% of requests that check venue data use information updated within the last 30 minutes.

3. Frontend design



Browser technology choices

Technology	Reason and trade-off
HTML / CSS / JavaScript	A small, static website with no framework to maintain. **A framework would make a larger website easier to organise and maintain.**
Cognito + PKCE	Handles sign-in; the APIs limit each traveller's access to their own trip. **Tokens stay in memory, so reloading restarts the browser session.**
DOM + semantic HTML	Shop descriptions are displayed as text to prevent embedded code from running. Labelled controls, keyboard navigation and clear status messages support accessibility.